

Contents

- [Useful LaTeX commands](#)
- [Insert formula](#)
- [Embed graphics](#)
- [Insert table](#)
- [List of abbreviations](#)
- [Bibliography](#)

- Installation:

<https://miktex.org/2.9/setup>

<https://texstudio.org/>

- Alternatively:

<https://overleaf.com> (next slide)

- Download the template:

Working with Overleaf

- Create account at <https://overleaf.com> (e.g. with your university email address via ORCDI)
- New Project -> Upload Project: Upload zip folder with the template
- Execute source code using the Recompile button
- Additional files can be added to the file overview (left) via drag & drop and can be selected and deleted there

- Create and display source code (F5)
Compile source code (F6)
-> Always compile source code twice to
update directories



Compiled
Source code

TOPIC OF THE WORK

TYPE OF WORK

presented by

AUTHOR

STREET/HOUSE NUMBER, POSTCODE CITY

Telephone: XXXX/XXXXXXX

MARTINA.MUSTERMANN@wirtschaft.uni-giessen.de

Bachelor of Science

6th Semester

MATRICULATION NUMBER

Advisor:

SUPERVISOR

CITY DDMMYYYY

```

\usepackage[TU-DD]{titlepage}
%Information title page
\university{\Huge Justus Liebig University Giessen}
\faculty{\LARGE Faculty of Business and Economics}
\chair{\Large Professorship in Economics VII for Statistics and Econometrics}
\professor{\Large Prof. Dr. Peter Winker}
%
\title{\Huge TOPIC OF THE WORK}[10pt]
}
\subject{TYPE OF WORK%
\large}
%
\author{\\[-10pt] \Large AUTHOR \\ STREET/HOUSE NUMBER, POSTCODE CITY \\ Telephone: XXXX/XXXXXXX \\ MARTINA.MUSTERMANN@wirtschaft.uni-giessen.de
\\[3pt]}
\discipline{Bachelor of Science \\ 6th Semester}
\matriculationnumber{MATRICULATION NUMBER}
%
\advisor{\Large SUPERVISOR}
\date{DD.MM.YYYY}
\place{CITY}

```

- Personalize front page information

Useful LaTeX commands

- Work is written between `\begin{document}` and `\end{document}`
- Commands are preceded by `\`
- Command `\clearpage` causes a new page to be started
- Line break is created with `\\`
- Paragraph is created with `\par` or with a blank line
- Chapters are created with `\section{}`
- Subchapters are created with `\subsection{}`
- Subsubchapters are created with `\subsubsection{}`

Useful LaTeX commands

- This puts the word in quotation marks: “example word” -> “Example word” (using double ‘)
- alternatively: “Example word” -> “Example word”
- Write text in italics: `\textit{}`
- Make text bold: `\textbf{}`
- Underline text: `\underline{}`
- Set footnotes: `\footnote{text of the footnote}`

Insert formula

- During a continuous text: in dollar signs $a+b=c$
- Set letters “mathematically”: a b α
- Formula symbols/mathematical symbols:
<https://www.codecogs.com/latex/eqneditor.php>

Display formula separated from text:

- For one-line equations:
$$a+b=c$$
- For multi-line equations:
$$a+b=c$$

Embed graphics

- The template is preset to allow **jpg files** to be included
- **jpg**. File should be saved in the same folder as the **tex file**
- Graphics are embedded between **\begin{figure}[h]** and **\end{figure}**
- With **\includegraphics [width=12cm]{name_of_the_file.pdf}** the file is uploaded

Embed graphics

```
\section{Figures}
\subsection{A figure}
\begin{figure}[h] % h important so that the image is placed "here"
\begin{center} %Image centered
\includegraphics[width=12cm]{bip_de.pdf} %Width of the image and name of the file
\caption{Gross domestic product of Germany at current prices, in billions of euros}
%Figure caption
\label{bip_de} %Reference for the text (see below)
\end{center}
\end{figure}
```

Figure \ref{bip_de} describes \lipsum[1]

Graphic has a width of 12cm
and is centered

3 Figures

3.1 A figure

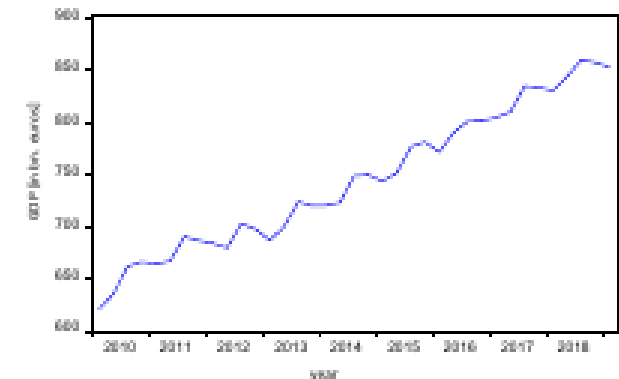


Figure 1: Gross domestic product of Germany at current prices, in billions of euros

Figure 1 describes Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut
purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida

Insert table

- Create a table between `\begin{table}[h]` and `\end{table}` so that the table can be labeled (`\caption{}`) and referenced (`\label{}`)
- If the table should be centered, create the table between `\begin{center}` and `\end{center}`
- Table is created between the commands `\begin{tabular}{|l|c|r|}` and `\end{tabular}`; `{}` defines the column properties
- Text of the first column of the table is left-aligned -> `l`, second column: centered -> `c`, third column: right-aligned -> `r`
- Contents of the individual cells are separated with `&`

Insert table

- Vertical line in the curly bracket: `|` ensures that columns are separated by a vertical line
- Horizontal lines are drawn with the command `\hline`
- If the horizontal line should only delimit cells in columns 2 to 3, use this command: `\cline{2-3}`
- Command `\hline \hline` ensures a small distance (by generating two horizontal lines) to the next line
- Individual column widths are specified by `p{}`:
`\begin{tabular}{|p{5cm}|p{4cm}|p{3.5cm}|}`

```

\begin{table}[h]
\begin{center} %if centered alignment is desired
\begin{tabular} {|c|c|c|c|} %Table with four columns is created, the text is centered and the columns
are separated by vertical lines
\hline %horizontal line is generated (to separate the rows)
Column heading 1 & Column heading 2 & Column heading 3 & Column heading 4 \\ \hline
example text & example text & example text & example text \\ \hline
example text & example text & example text & example text \\ \hline
example text & example text & example text & example text \\ \cline{2-3} %horizontal line is only
generated for cells of column 2 to 3
example text & example text & example text & example text \\ \hline
example text & example text & example text & example text \\ \hline \hline
example text & example text & example text & example text \\ \hline
\end{tabular}
\end{center}
\caption{Example table} %Table caption
\label{tab_a}
\end{table}

```

Column heading 1	Column heading 2	Column heading 3	Column heading 4
example text	example text	example text	example text
example text	example text	example text	example text
example text	example text	example text	example text
example text	example text	example text	example text
example text	example text	example text	example text
example text	example text	example text	example text

Table 1: Example table

Insert table

- Very useful website that provides assistance in creating tables:

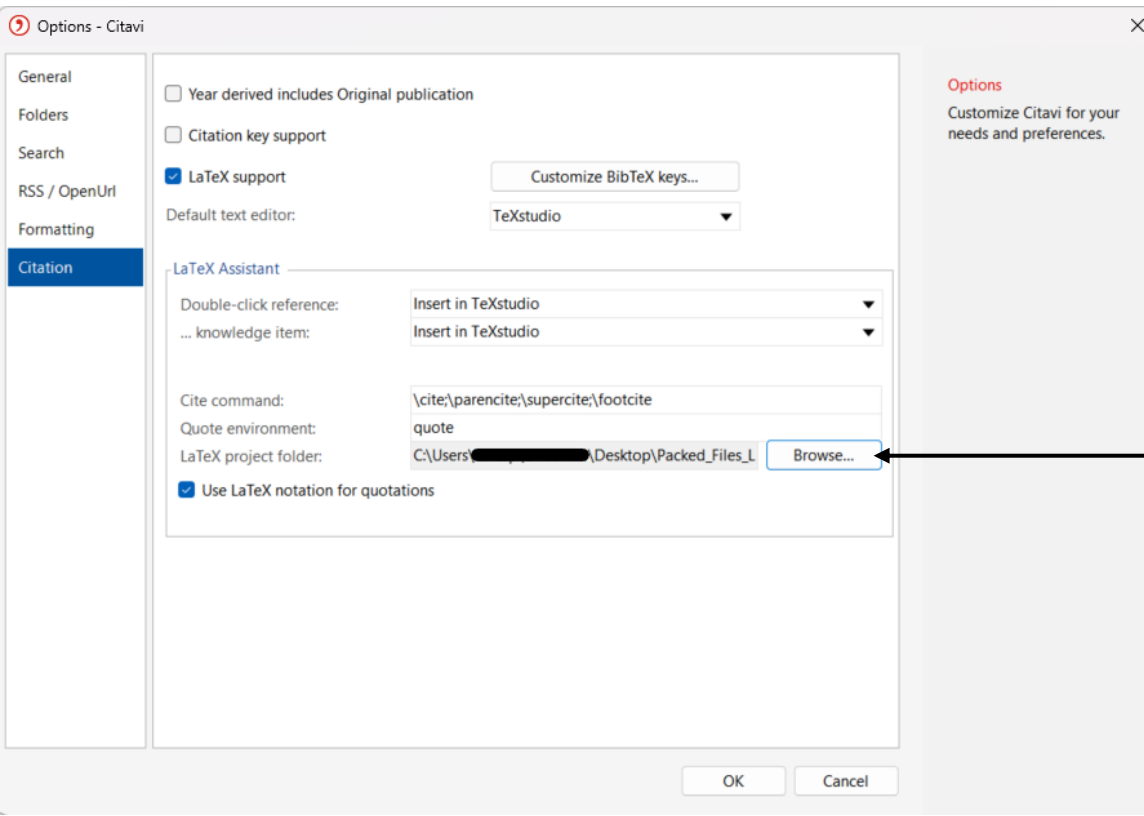
<https://www.tablesgenerator.com/>

List of abbreviations

- To use, import the package with `\usepackage{acronym}` at the beginning of the document
- created between `\begin{acronym}[List of abbreviations]` and `\end{acronym}`
- Further abbreviations are added with `\acro{ }{ }`
- For example: `\acro{GDP}{Gross Domestic Product}`
- In the body text use `\ac{ }`, i.e. therefore use `\ac{GDP}`
- `\acs{GDP}` causes abbreviation to be written in short form
- `\acl{GDP}` causes abbreviation to be written in long form

Bibliography

- Create a new project in Citavi and import the desired sources
- Tools -> Options -> Citation: Enable LaTeX support



Specify project path

Bibliography

- Export literature sources:

File -> Export -> Export -> Export title in BibTeX format

- Specify the name and location of the export file -> Name the export file as “**literature**”

```

\DeclareFieldFormat
[article,inbook,incollection,inproceedings,patent,thesis,unpublished]
{title}{#1} % Suppresses quotation marks for articles in the bibliography
\renewbibmacro{in:}{\ifentrytype{article}{}{\printtext{\bibstring{in}\intitlepunct}}}
\renewcommand*{\newunitpunct}{\addcomma\space}
%%
%%
\bibliography{literature}

%%
%%
%Short citation style
%%
\renewcommand{\postnotedelim}{: } % Separator between citation and page number
\DeclareFieldFormat{postnote}{#1}
\DeclareFieldFormat{multipostnote}{#1} % suppresses "p." before the page number in the citation

\usepackage{setspace}
\setstretch{1.5} % Package to set line spacing to REAL 1.5!

\usepackage{array} % Extended table environment
\usepackage{calc}

```

It is important that the export file is named "literature." The export file name must always match the name in the curly brackets.

Bibliography

- BibTex file is now in the project folder
- Citavi: Citation -> LaTeX Assistant -> Select Source
- Adjust citation with the commands:

`\parencite[cf.][page number]{source name}`

`\parencite[page number]{source name}`

`\textcite[page number]{source name}`

`\subsection{Citation}`

Citations are entered as follows:\\

`\parencite[cf.][112-114]{Winker,2017}` \\%Citation in parentheses, cf., page number

`\parencite[50-54]{Lutkepohl,2006}`\\ %Citation in parentheses, page number

`\textcite[470]{Makridakis,2000}`\\ %Citation without parentheses, page number

`\textcite{Makridakis,2000}`\\ %Citation without parentheses

2.3 Citation

Citations are entered as follows:

(cf. Winker 2017: 112-114)

(Lütkepohl 2006: 50-54)

Makridakis and Hibon (2000: 470)

Makridakis and Hibon (2000)